

对所有点 $\text{rand} \cdot \rightarrow \begin{cases} \uparrow \\ \downarrow \end{cases}$ 搬运 $\rightarrow H_1$

随机反转 - $\phi \rightarrow H_1$ 计算 $\Delta H \begin{cases} \Delta H < 0 \Rightarrow \text{反转} \\ \Delta H > 0 \end{cases}$

Metropolis 判据

$$p = \exp\left(-\frac{\Delta H}{kT}\right)$$

$$\text{rand} \in [0,1] \rightarrow \begin{cases} \text{rand} < p \\ \text{rand} > p \end{cases}$$

$$\begin{cases} \mathbf{v} \cdot \vec{e} = 0 \end{cases} \Rightarrow \text{叠加原理}$$

$$\begin{cases} \oint_L \vec{E} \cdot d\vec{l} = 0 \end{cases} \Rightarrow \gamma = \frac{kq\sigma}{r^2} \quad \gamma$$

01-02

对库仑定律的讨论：并己

一、电磁场的相对论变换

$$\textcircled{2} \text{ 恒力矢量 } \vec{F} := \frac{d\vec{p}}{dt} = (F_x, F_y, F_z, \frac{1}{c} \frac{dE}{dt}) \quad \vec{F} = \frac{d\vec{p}}{dt} \quad F_4 = \frac{dE}{dt} = \frac{d}{dt} \left(\frac{1}{2} m_0 v^2 \right)$$

是 Lorentz 不变量

$$\text{满足} \quad \left\{ \begin{array}{l} F'_x = \gamma(F_x + \beta F_z) \\ F'_y = F_y \\ F'_z = F_z \\ F'_4 = \gamma(F_4 + \beta F_x) \end{array} \right. \quad \left\{ \begin{array}{l} F_x = \gamma(F'_x - \beta F'_z) \\ F_y = F'_y \\ F_z = F'_z \\ F_4 = \gamma(F'_4 + \beta F'_x) \end{array} \right.$$

电磁场中： $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$

$$\left\{ \begin{array}{l} F_x = q(E_x + v_y B_z - v_z B_y) \\ F_y = q(E_y + v_z B_x - v_x B_z) \\ F_z = q(E_z + v_x B_y - v_y B_x) \\ \frac{1}{c} \frac{dE}{dt} = \frac{1}{c} q(v_x E_x + v_y E_y + v_z E_z) \quad (\text{洛伦兹力与速度} \perp) \end{array} \right.$$

$$\left\{ \begin{array}{l} F_x = q \left(\frac{1}{c} u_z E_x + u_y B_z - u_z B_y \right) \\ F_y = q \left(\frac{1}{c} u_z E_y + u_x B_z - u_x B_y \right) \\ F_z = q \left(\frac{1}{c} u_z E_z + u_x B_y - u_y B_x \right) \\ F_4 = \frac{1}{c} q (u_x E_x + u_y E_y + u_z E_z) \end{array} \right. \quad u_x = \frac{1}{c} \frac{dx}{dt}$$

Lorentz 变换

$$F'_x = \gamma(F_x + \beta F_z)$$

$$= \gamma q \left[\left(\frac{1}{c} u_z E_x + u_y B_z - u_z B_y \right) + \beta \frac{1}{c} (u_x E_x + u_y E_y + u_z E_z) \right]$$

$$u_x = \gamma(u'_x + \beta u'_z), \quad u_y = u'_y, \quad u_z = u'_z, \quad u_4 = \gamma(u'_4 - \beta u'_x)$$

$$= - \frac{1}{c} q E_x \cdot \gamma(u'_x + \beta u'_z) + \left(\gamma q B_z - \gamma \beta \frac{q}{c} E_y \right) u'_y - \left(\gamma q B_y + \gamma \beta \frac{q}{c} E_z \right) u'_z$$

$$= - \left(\beta \frac{q}{c} E_x \gamma^2 - \frac{\gamma^2}{c} \beta E_x \right) u'_x + \left(\gamma q B_z - \gamma \beta \frac{q}{c} E_y \right) u'_y - \left(\gamma q B_y + \gamma \beta \frac{q}{c} E_z \right) u'_z$$

$$= \partial_y (B_z - \frac{v}{c} E_y) u'_y - \partial_y (B_y + \frac{v}{c} E_z) u'_z - i \partial^2 \frac{q}{c} (1-\beta^2) z_x u'_z$$

$$= q \left(\frac{1}{c} u'_z E'_x + u'_y B'_z - u'_z B'_y \right) \quad \text{同前形式一致}$$

$$\rightarrow B'_z = \partial (B_z - \frac{v}{c} E_y)$$

$$B'_y = \partial (B_y + \frac{v}{c} E_z)$$

$$E'_x = \partial^2 (1-\beta^2) z_x = z_x$$

利用其他方程，类似有

$$E'_x = z_x$$

$$B'_x = B_x$$

$$E'_y = \partial (E_y - c\beta B_z)$$

$$B'_y = \partial (B_y + \frac{v}{c} E_z)$$

$$E'_z = \partial (E_z + c\beta B_y)$$

$$B'_z = \partial (B_z - \frac{v}{c} E_y)$$

二、运动点电荷的电场：

$$K' \text{ 系中静止点电荷电场为: } \vec{E}' = \frac{q}{4\pi\epsilon_0} \frac{\vec{r}'}{r'^3}$$

$$\text{磁场 } B'_x = B'_y = B'_z = 0$$

K' 相对 K 以 v 沿 x 轴正向运动

$$E_x = E'_x = \frac{q}{4\pi\epsilon_0} \frac{x'}{r'^3} \quad E_y = 0 \quad E_z = 0$$

$$x = x' - vt, \quad y' = y, \quad z' = z$$

$$\Rightarrow E_x = \frac{q}{4\pi\epsilon_0} \frac{\partial(x-vt)}{[x^2(x-vt)^2 + y^2 + z^2]^{3/2}}$$

$$E_y = \frac{q}{4\pi\epsilon_0} \frac{\partial y}{[x^2(x-vt)^2 + y^2 + z^2]^{3/2}}$$

$$E_z = \frac{q}{4\pi\epsilon_0} \frac{\partial z}{[x^2(x-vt)^2 + y^2 + z^2]^{3/2}}$$

$$E = 0 \quad , \quad \frac{E_x}{R} = \frac{E_y}{y} = \frac{E_z}{z}$$

$$\begin{aligned}
 E^2 &= E_x^2 + E_y^2 + E_z^2 = \left(\frac{q}{4\pi\epsilon_0} \right)^2 \frac{\partial^2 (x^2 + y^2 + z^2)}{(\partial^2 x^2 + y^2 + z^2)^3} \\
 &= \left(\frac{q}{4\pi\epsilon_0} \right)^2 \frac{1}{x^2 + y^2 + z^2} \left\{ \partial^2 - \frac{(\partial^2 - 1)(y^2 + z^2)}{x^2 + y^2 + z^2} \right\}^2 \\
 &= \left(\frac{q}{4\pi\epsilon_0} \right)^2 \frac{1}{(x^2 + y^2 + z^2)^3} \left\{ 1 - \frac{(\partial^2 - 1)(y^2 + z^2)}{x^2 + y^2 + z^2} \right\}^2
 \end{aligned}$$

$\partial^2 - 1 = \frac{1}{1 - \beta^2} - 1 = \frac{\beta^2}{1 - \beta^2}$
 $\frac{\partial^2 - 1}{\partial^2} = \beta^2$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \cdot \frac{(1 - \beta^2)}{[1 - \beta^2 \sin^2 \theta]^{3/2}} \quad \text{其中 } \theta = \langle \vec{v}, \vec{r} \rangle$$

亦可写为

$$E = E_0 \frac{1 - \beta^2}{[1 - \beta^2 + \beta^2 \cos^2 \theta]^{3/2}} = E_0 \frac{1 - \beta^2}{[1 - \beta^2 + (\frac{\vec{v} \cdot \vec{r}}{cr})^2]^{3/2}} \quad \text{为所求形式}$$

三、运动电荷的磁场

$$B_x = 0$$

$$B_y = -\frac{1}{c} \frac{\partial}{\partial t} E_z' = -\frac{\beta}{c} E_z$$

$$B_z = \frac{1}{c} \frac{\partial}{\partial t} E_y' = \frac{\beta}{c} E_y$$

$$\Rightarrow \vec{B} = \frac{1}{c^2} \vec{v} \times \vec{E}$$

考虑磁力的时候 ✓

01-03 练习 2 例 1.1.11

$$F_r = 2 \cdot \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2 + (\frac{d}{2})^2} \cdot \frac{r}{(r^2 + (\frac{d}{2})^2)^{3/2}}$$

$$= \frac{Qq}{2\pi\epsilon_0} \cdot \frac{r}{[r^2 + (\frac{d}{2})^2]^{3/2}} = \alpha \cdot \frac{r}{[r^2 + (\frac{d}{2})^2]^{3/2}}$$

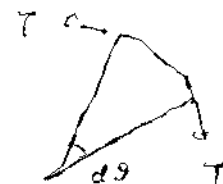
$$r \ll d \rightarrow F_r = \alpha \cdot r \cdot \frac{1}{(\frac{d}{2})^3} = \frac{8\alpha}{d^3} r = m\ddot{r}$$

$$W = \int \frac{d^2}{m\dot{r}^2} = \int -\frac{4Qq}{2\pi\epsilon_0 d^3} \cdot$$



01-03 练习 3 例 1.1.19

$$\frac{q\lambda R Q}{4\pi\epsilon_0 R^2} = 2T \sin(\frac{\theta}{2}) \approx T \cdot \theta \quad \text{其中 } \theta = \frac{Q}{2\lambda R}$$



$$\text{得 } T = \frac{qQ}{8\pi^2\epsilon_0 R^2}$$

01-03 练习 4

$$\frac{e^2}{4\pi\epsilon_0} \frac{1}{r^2} = m_e \omega^2 r \quad \text{得 } \omega = \sqrt{\frac{e^2}{4\pi\epsilon_0 r^3 m_e}}$$

$$L = m_e \omega r^2 = \sqrt{\frac{m_e e^2}{4\pi\epsilon_0}} \quad \overline{r} = n \cdot \bar{a}$$

$$\Rightarrow r_n = n^2 \bar{a} \quad \bar{a} = \sqrt{\frac{4\pi\epsilon_0}{m_e e^2}} \cdot \hbar$$

01-03 练习 5

$$F = \frac{e^2}{4\pi\epsilon_0 r^2}$$

$$r_1 = \frac{m_p}{m_e + m_p} r, \quad r_2 = \frac{m_e}{m_e + m_p} r$$

$$F_1 = \frac{e^2}{4\pi\epsilon_0 r_1^2} \cdot \left(\frac{m_p}{m_e + m_p}\right)^2 \quad F_2 = \frac{e^2}{4\pi\epsilon_0 r_2^2} \cdot \left(\frac{m_e}{m_e + m_p}\right)^2$$

$$= -m_e \frac{d^2 r_i}{dt^2}$$

$$= -m_p \frac{d^2 r_i}{dt^2}$$

$$\frac{1}{2} k = \frac{e^2}{4\pi\epsilon_0(m_e+m_p)^2}$$

$$- \frac{k m_p^2}{r^3 m_e} = \ddot{r}_i = \frac{dr_i}{dt} \cdot \frac{dr_i}{dr_i} = \dot{r}_i \frac{dr_i}{dr_i}$$

$$\Rightarrow \frac{k m_p^2}{m_e} \left(\frac{1}{r_i} - \frac{1}{r_{i0}} \right) = \frac{1}{2} \dot{r}_i^2 - 0$$

$$\text{解 } \dot{r}_i = -m_p \sqrt{\frac{2k}{m_e}} \left| \frac{1}{r_i} - \frac{1}{r_{i0}} \right| \quad (\text{因为 } \dot{r}_i < 0, r_i < 0)$$

$$\frac{dr_i}{\left| \frac{1}{r_i} - \frac{1}{r_{i0}} \right|} = -m_p \sqrt{\frac{2k}{m_e}} dr_i$$

$$\int \frac{dx}{\left| \frac{1}{x} - \frac{1}{a} \right|} = \int \frac{ax}{a-x} dx$$

$$u = a-x \quad x = a-u^r \quad dx = -du$$

$$\int \frac{ax}{a-x} dx = \int \frac{a(a-u)}{u} - 2u du = -2a \int \frac{1}{u} du$$

$$u = a \sin t$$

$$= -2a \int \frac{1}{a \cos t} \cdot a \cos t dt = -2a^{3/2} \int \frac{1}{1 + \cos(2t)} dt$$

$$= -a^{3/2} \left[t + \frac{1}{2} \sin(2t) \right] \quad \sin(2t) = 2 \sin t \cos t = 2 \cdot \sqrt{\frac{a-x}{a}} \cdot \sqrt{\frac{x}{a}}$$

$$= -a^{3/2} \left(\arcsin \left(\sqrt{\frac{a-x}{a}} \right) + \frac{(x(a-x))}{a} \right)$$

$$\frac{dr_i}{\left| \frac{1}{r_i} - \frac{1}{r_{i0}} \right|} = -r_{i0}^{3/2} \left[\arcsin \left(\sqrt{\frac{r_{i0}-r_i}{r_{i0}}} \right) + \frac{(r_{i0}-r_i)r_i}{r_{i0}} \right] \Bigg|_{r_i=r_{i0}}^{r_i=r_i}$$

$$= r_{i0}^{3/2} \left[\arcsin \left(\sqrt{\frac{r_{i0}-r_i}{r_{i0}}} \right) + \frac{(r_{i0}-r_i)r_i}{r_{i0}} \right]$$

$$t_0 = \frac{\sqrt{m_e}}{2k} \cdot \frac{1}{m_p} \cdot r_{i0}^{3/2} \left[\arcsin \left(\sqrt{\frac{r_{i0}-r_i}{r_{i0}}} \right) + \frac{(r_{i0}-r_i)r_i}{r_{i0}} \right] \Bigg|_{r_i=0}$$

$$= \frac{(4\pi\epsilon_0 m_e(m_e+m_p)^2)}{2e^2} \frac{1}{m_p} r_{i0}^{3/2} \cdot \frac{\pi}{2} = \frac{(2\pi\epsilon_0 m_e}{e m_p} \cdot r_{i0}^{3/2} \cdot \frac{\pi}{2}$$

$$= \frac{(2\pi\epsilon_0 m_e}{e m_p} \cdot r_{i0}^{3/2} \cdot \frac{\pi}{2} \quad r_{i0} = 1 \cdot \frac{m_p}{m_e+m_p}$$

$$= \left(\frac{\lambda^3 \epsilon_0 m_e m_p}{2(m_e + m_p)} \right) \cdot \frac{1}{e} = \left(\frac{\lambda \epsilon_0}{2e^2 \lambda} \right) \sqrt{\frac{1}{\mu}} \quad \text{课例上为1}$$

01-03 练习6 例 1.1.20

设 Q, λ_0, R

$$\lambda_1 = \frac{Q}{2\pi R}$$

求 x 处电场元电荷环上作用力

$$dF(x) = \frac{\lambda_0 dx}{4\pi \epsilon_0} \cdot \frac{1}{R^2 + x^2} \int_0^{2\pi} \lambda_1 R d\theta$$

$$= \frac{1}{2\epsilon_0} \frac{\lambda_0 Q}{R^2 + x^2} dx$$

$$F = \int_0^\infty dF(x)$$

$$= \frac{\lambda_0 Q}{2\epsilon_0} \int_0^\infty \frac{dx}{R^2 + x^2} = \frac{\lambda_0 Q}{2\epsilon_0 R} \int_0^\infty \frac{d\frac{x}{R}}{1 + (\frac{x}{R})^2}$$

$$= \frac{\lambda_0 Q}{2\epsilon_0 R} (\arctan \infty - \arctan 0) = \frac{\lambda_0 Q}{4\epsilon_0 R}$$

01-03 练习7

$$\begin{cases} m(\ddot{r} - r\dot{\theta}^2) = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \\ m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = 0 \end{cases}$$

$$\dot{h} = r^2 \dot{\theta} \quad dh = 2r\dot{r}\dot{\theta} + r^2\ddot{\theta} = r(2\dot{r}\dot{\theta} + r\ddot{\theta}) = 0$$

$$\text{故 } h = \text{const.} \quad \dot{\theta} = \frac{h}{r^2}$$

$$E = \frac{1}{2} m v^2 - \frac{qQ}{4\pi\epsilon_0 r} = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) - \frac{1}{4\pi\epsilon_0} \frac{qQ}{r}$$

$$\text{代入 } r, \dot{\theta} = \frac{h}{r^2} \quad \dot{r} = \frac{dr}{dt}, \quad \frac{dr}{dt} = \frac{dr}{d\varphi} \cdot \frac{d\varphi}{dt} = \frac{dr}{d\varphi} \cdot \frac{h}{r^2}$$

$$= \frac{1}{2} m \left[\left(\frac{dr}{d\varphi} \right)^2 \left(\frac{h}{r^2} \right)^2 + r^2 \cdot \frac{h^2}{r^4} \right] - \frac{1}{4\pi\epsilon_0} \frac{qQ}{r}$$

$$= \frac{1}{2} m \left[\left(\frac{dr}{d\varphi} \right)^2 \frac{h^2}{r^4} + \frac{h^2}{r^2} \right] - \frac{qQ}{4\pi\epsilon_0} \frac{1}{r}$$

$$\frac{1}{r} = u$$

$$\frac{dr}{d\varphi} = \frac{dr}{du} \cdot \frac{du}{d\varphi} = -\frac{1}{u^2} \frac{du}{d\varphi}$$

$$E = \frac{1}{2} m \left[\dot{u}^2 \left(\frac{du}{d\theta} \right)^2 + u^4 \right] - \frac{qQ}{4\pi\epsilon_0} u$$

$$= \frac{1}{2} m \dot{u}^2 \left[\left(\frac{du}{d\theta} \right)^2 + u^2 \right] - \frac{qQ}{4\pi\epsilon_0} u$$

$$\frac{1}{2} m \dot{u}^2 \left[\left(\frac{du}{d\theta} \right)^2 + u^2 \right] - \frac{qQ}{4\pi\epsilon_0} u - E = 0$$

上式两边求导.

$$m \dot{u}^2 \left(\frac{du}{d\theta} \frac{d^2 u}{d\theta^2} + u \frac{du}{d\theta} \right) - \frac{qQ}{4\pi\epsilon_0} \frac{du}{d\theta} = 0$$

$$\Rightarrow \frac{d^2 u}{d\theta^2} + u - \frac{qQ}{4\pi\epsilon_0 m \dot{u}^2} = 0$$

$$\text{解得 } u = A \cos \theta + B \sin \theta + \frac{qQ}{4\pi\epsilon_0 m \dot{u}^2}$$

$$r = \frac{1}{u} = \frac{1}{1 + \frac{qQ}{4\pi\epsilon_0 m \dot{u}^2} \cos(\theta + \varphi)} = \frac{p}{1 + e \cos \varphi}$$

为极坐标下的圆锥曲线方程.

01-03 练习 8

$$F(x) = 2 \cdot \frac{1}{4\pi\epsilon_0} \cdot \frac{q_0 Q}{(l^2 + x^2)^{3/2}} \cdot x$$

$$= \frac{q_0 Q}{2\pi\epsilon_0} \frac{x}{(l^2 + x^2)^{3/2}}$$

$$\frac{dF}{dx} = \frac{q_0 Q}{2\pi\epsilon_0} \frac{1}{(l^2 + x^2)^{3/2}} \left[1 \cdot (l^2 + x^2)^{3/2} - x \cdot \frac{3}{2} (l^2 + x^2)^{1/2} \cdot 2x \right]$$

$$= \frac{q_0 Q}{2\pi\epsilon_0 (l^2 + x^2)^{3/2}} (l^2 + x^2 - 3x^2) = \frac{q_0 Q (l^2 - 2x^2)}{2\pi\epsilon_0 (l^2 + x^2)^{3/2}}$$

$$\Rightarrow x = \frac{\sqrt{2}}{2} l \text{ 处}$$

26

0.

01-06 电偶极子

1. 轴向电场

$$\begin{aligned}
 E &= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{(r-\frac{l}{2})^2} - \frac{1}{(r+\frac{l}{2})^2} \right] \\
 &\approx \frac{q}{4\pi\epsilon_0} \frac{(r+\frac{l}{2} + r-\frac{l}{2}) (r+\frac{l}{2} - r+\frac{l}{2})}{r^4} \\
 &= \frac{q}{4\pi\epsilon_0} \frac{2l}{r^3} = \frac{2p}{4\pi\epsilon_0 r^3}
 \end{aligned}$$

2. 角向电场

$$E = \frac{q}{4\pi\epsilon_0} \frac{2}{(r^2 + \frac{l^2}{4})^{3/2}} \approx \frac{p}{4\pi\epsilon_0 r^3}$$

3. 任意 (r, θ) 处电场

两个电荷极化叠加

$$\begin{aligned}
 E &= \frac{2p_r}{4\pi\epsilon_0 r^3} \hat{r} + \frac{p_\theta}{4\pi\epsilon_0 r^3} \hat{\theta} \\
 &= \frac{p}{4\pi\epsilon_0 r^3} (2\cos\theta \hat{r} + \sin\theta \hat{\theta})
 \end{aligned}$$

$$\frac{\vec{p} \cdot \vec{r}}{|\vec{p}| |\vec{r}|} = \cos\theta \quad \vec{p} = p \cos\theta \hat{r} - p \sin\theta \hat{\theta}$$

$$= \frac{p}{4\pi\epsilon_0 r^3} [3\cos\theta \hat{r} - (\cos\theta \hat{r} - \sin\theta \hat{\theta})]$$

$$= \frac{1}{4\pi\epsilon_0} \left[\frac{3\vec{p} \cdot \vec{r}}{r^5} \vec{r} - \frac{\vec{p}}{r^3} \right]$$

$$-\frac{1}{r^3} \vec{p} \cos\theta + \frac{1}{r^3} \vec{p} \sin\theta$$

01-06 同极子线电场

在 xoy 平面

$$E = \frac{1}{4\pi\epsilon_0} \left[\frac{-2q}{r^2} + \frac{2q}{(r^2+l^2)} \right]$$

$$= -\frac{2q}{4\pi\epsilon_0} \frac{l^2}{(r^2+l^2)r^3}$$

在 z 轴上

$$E = \frac{1}{4\pi\epsilon_0} \left[\frac{-2q}{r^2} + \frac{q}{(r-l)^2} + \frac{q}{(r+l)^2} \right]$$

$$\approx \frac{q}{4\pi\epsilon_0} \frac{2l^2}{r^4}$$

01-06 电偶极子在外场受力

$$\tau = 2ql \frac{1}{2} \sin \theta$$

$$= \vec{p} \times \vec{E}$$

$$\begin{matrix} & z \\ q & 0 \\ & l \\ -q & 0 \\ & -l \\ q & 0 \\ & x \end{matrix}$$

$$\begin{matrix} \vec{z} \\ \vec{y} \\ \vec{x} \\ 0 \end{matrix} \quad \frac{2q}{r^3}$$

01-08 电场强度计算.

$$(1) dE(x) = \frac{1}{4\pi\epsilon_0} \frac{2\lambda dx}{(r^2+x^2)^{3/2}} r$$

$$E = \int_0^L dE(x)$$

$$= \frac{2}{4\pi\epsilon_0} \frac{q}{Lr} \int_0^L \frac{dx}{(r^2+x^2)^{3/2}}$$

$$= \frac{q}{4\pi\epsilon_0 Lr} \int_0^{\frac{1}{r}} \frac{dk}{(1+k^2)^{3/2}}$$

$$\frac{1}{3} k = \tan \theta \quad dk = \sec^2 \theta$$

$$\int_0^{\frac{1}{r}} \frac{dk}{(1+k^2)^{3/2}} = \int_0^{\theta_0} \frac{\sec^2 \theta}{\sec^3 \theta} d\theta = \int_0^{\theta_0} \cos \theta d\theta = \sin(\arctan(\frac{L}{r}))$$

$$= \frac{1}{\sqrt{L^2+r^2}}$$

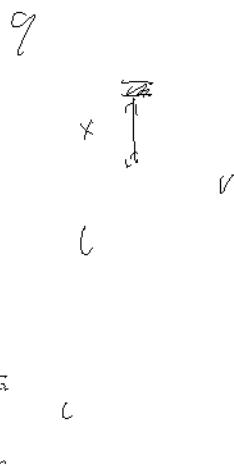
$$E(r) = \frac{q}{4\pi\epsilon_0} \frac{1}{r\sqrt{L^2+r^2}}$$

讨论:

无限长直导线: $L \rightarrow \infty$

$$E(\infty) = \frac{q}{L} = \frac{1}{4\pi\epsilon_0 r} \lambda = \frac{\lambda}{2\pi\epsilon_0 r} \quad \lambda = \frac{q}{L}$$

任意位置? :



2. 圆环



$$E(x) = \frac{Q}{4\pi\epsilon_0} \frac{x}{(a^2+x^2)^{3/2}}$$

讨论: $x \rightarrow \infty$ 时,

退化为点电荷, $E(x) = \frac{Q}{4\pi\epsilon_0 x^2}$

任意处 E?

3. 圆筒



$$dE(x, r) = \frac{1}{4\pi\epsilon_0} \cdot \frac{L 2\pi r dr}{(r^2+x^2)^{3/2}} x$$

$$= \frac{Lx}{2\epsilon_0} \frac{r dr}{(r^2+x^2)^{3/2}}$$

$$E(x, r) = \int_0^a dE(x, r)$$

$$= \frac{Lx}{2\epsilon_0} \int_0^a \frac{r dr}{(r^2+x^2)^{3/2}}$$

$$= \frac{Lx}{4\epsilon_0} \left[-2(r^2+x^2)^{-1/2} \right]_0^a$$

$$= \frac{Lx}{2\epsilon_0} \left(\frac{1}{x} - \frac{1}{\sqrt{a^2+x^2}} \right)$$

$$= \frac{Q}{2\pi\epsilon_0 a} \left(1 - \frac{x}{\sqrt{a^2+x^2}} \right)$$

$$L = \frac{Q}{\lambda}$$

讨论: $a \rightarrow \infty$ 时

无穷大带电平面: $E = \frac{\sigma}{2\epsilon_0}$

$x \rightarrow \infty$ 时

$$\frac{x}{\sqrt{a^2+x^2}} = \frac{1}{\sqrt{1+(\frac{a}{x})^2}} = 1 - \frac{1}{2} \left(\frac{a}{x}\right)^2 + \dots$$

可以视为点电荷吗?

01-08

例 2.14

$$dE_x(x) = - \frac{\lambda dx}{4\pi\epsilon_0} \frac{x}{(x^2+y^2)^{3/2}}$$

$$E_x = \int_0^\infty dE_x(x)$$

$$= - \frac{\lambda}{4\pi\epsilon_0} \int_0^\infty \frac{x dx}{(x^2+y^2)^{3/2}}$$

$$= - \frac{\lambda}{4\pi\epsilon_0} \left[\frac{1}{2} (-2) (x^2+y^2)^{-1/2} \right]_0^\infty$$

$$= - \frac{\lambda}{4\pi\epsilon_0} \cdot \frac{1}{y}$$

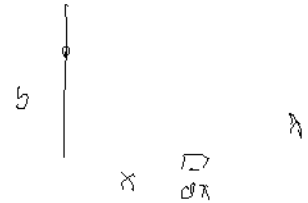
$$dE_y(x) = \frac{\lambda dx}{4\pi\epsilon_0} \frac{y}{(x^2+y^2)^{3/2}}$$

$$E_y(x) = \int_0^\infty dE_y(x)$$

$$= \frac{\lambda y}{4\pi\epsilon_0} \int_0^\infty \frac{dx}{(x^2+y^2)^{3/2}}$$

$$= \frac{\lambda}{4\pi\epsilon_0 y} \int_0^\infty \frac{d(\frac{x}{y})}{[1+(\frac{x}{y})^2]^{3/2}}$$

$$= \frac{\lambda}{4\pi\epsilon_0 y} \sin(\arctan(\frac{x}{y})) = \frac{\lambda}{4\pi\epsilon_0 y}$$

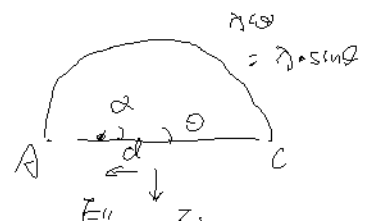


01-08

例 2.19

$$dE_{||}(\theta) = \int_0^\pi \frac{\lambda R d\theta}{4\pi\epsilon_0 R^2} \cos\theta$$

$$= \frac{\lambda_0}{4\pi\epsilon_0 R} \int_0^\pi \sin\theta \cos\theta d\theta = 0$$



例 2: 求 $\oint_C \frac{1}{z} dz$

$$dE_1(\theta) = \int_0^{2\pi} \frac{\lambda R d\theta}{4\pi \epsilon_0} \cdot \frac{\cos \theta}{R^2 + d^2 + 2Rd \cos \theta}$$

$$d\cos \theta = \frac{d + R \cos \theta}{R^2 + d^2 + 2Rd \cos \theta}$$

$$= \int_0^{2\pi} \frac{\lambda R}{4\pi \epsilon_0} \cdot \frac{\sin \theta (R \cos \theta + d)}{(R^2 + d^2 + 2Rd \cos \theta)^{3/2}} d\theta$$

$$\int_0^{2\pi} \frac{\sin \theta \cos \theta d\theta}{(R^2 + d^2 + 2Rd \cos \theta)^{3/2}}$$

$$\int \frac{x dx}{x \sqrt{x}} = -\frac{2(bx + a)}{9 \sqrt{x}}$$

$$= - \int_0^{2\pi} \frac{\cos \theta d(\cos \theta)}{(R^2 + d^2 + 2Rd \cos \theta)^{3/2}}$$

$$X = a + bx + cx^2 \quad g = 4ac - b^2$$

$$= - \left[\frac{2 \cdot 2Rd \cos \theta + 2(R^2 + d^2)}{(2Rd)^2 (R^2 + d^2 + 2Rd \cos \theta)^{3/2}} \right] \Big|_0^{2\pi}$$

$$= \frac{1}{(Rd)^2} \left(\frac{R^2 + d^2 + Rd}{R + d} - \frac{R^2 + d^2 - Rd}{R - d} \right) = \frac{1}{(Rd)^2 (R^2 - d^2)} [(R^2 + d^2 + Rd)(R - d) - (R^2 + d^2 - Rd)(R + d)]$$

$$= \frac{1}{(Rd)^2 (R^2 - d^2)} [-d(R^2 + d^2) + R^2 d + R^2 d - d(R^2 + d^2)] = -\frac{2d}{R^2 (R^2 - d^2)}$$

$$\int_0^{2\pi} \frac{\sin \theta d\theta}{(R^2 + d^2 + 2Rd \cos \theta)^{3/2}}$$

$$= - \int_0^{2\pi} \frac{d(\cos \theta)}{(R^2 + d^2 + 2Rd \cos \theta)^{3/2}}$$

$$\int \frac{dx}{x \sqrt{x}} = \frac{2(ax + b)}{9 \sqrt{x}}$$

$$= - \frac{2 [2Rd]}{(2Rd)^2 (R^2 + d^2 + 2Rd \cos \theta)^{3/2}} \Big|_0^{2\pi} = \frac{1}{Rd} \left(\frac{1}{R-d} - \frac{1}{R+d} \right)$$

$$= \frac{2}{R(R^2 - d^2)}$$

$$\int_0^{2\pi} \frac{\lambda R}{4\pi \epsilon_0} \cdot \frac{\sin \theta (R \cos \theta + d)}{(R^2 + d^2 + 2Rd \cos \theta)^{3/2}} d\theta$$

$$= \frac{\lambda R}{4\pi \epsilon_0} \int_0^{2\pi} \frac{1}{(R^2 + d^2 + 2Rd \cos \theta)^{3/2}} (R \sin \theta \cos \theta + d \sin \theta) d\theta$$

$$= \frac{\lambda R}{4\pi \epsilon_0} \left[R \cdot \frac{-2d}{R^2 (R^2 - d^2)} + d \cdot \frac{2}{R(R^2 - d^2)} \right] = 0$$

01-08

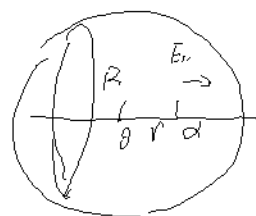
137 1.2.25

$$dE_r(\theta) = \frac{6 \cdot 2\pi R^2 \sin\theta d\theta}{4\pi\epsilon_0} \cdot \frac{\cos\theta}{R^2 + r^2 + 2Rr\cos\theta}$$

$$\cos\theta = \frac{R\cos\theta + r}{R^2 + r^2 + 2Rr\cos\theta}$$

$$E_r = \int_0^\pi dE_r(\theta)$$

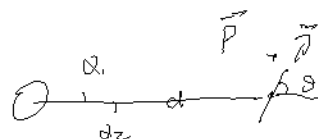
$$= \frac{6R^2}{2\epsilon_0} \int_0^\pi \frac{\sin\theta (R\cos\theta + r)}{(R^2 + r^2 + 2Rr\cos\theta)^{3/2}} d\theta = 0$$



01-08

137 1

Q



$$F_x = \frac{qQ}{4\pi\epsilon_0} \cdot \frac{\cos\theta}{d^2 + \frac{1}{4}z^2 + d\cos\theta} = \frac{qQ}{4\pi\epsilon_0} \frac{d + \frac{1}{2}z\cos\theta}{[d^2 + \frac{1}{4}z^2 + d\cos\theta]^{3/2}}$$

$$F_y = \frac{qQ}{4\pi\epsilon_0} \frac{\frac{1}{2}z\sin\theta}{[d^2 + \frac{1}{4}z^2 + d\cos\theta]^{3/2}}$$

$$F_x = -\frac{qQ}{4\pi\epsilon_0} \frac{d - \frac{1}{2}z\cos\theta}{[d^2 + \frac{1}{4}z^2 - d\cos\theta]^{3/2}}$$

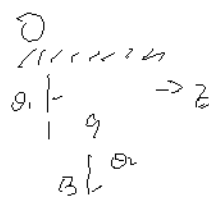
$$F_y = -\frac{qQ}{4\pi\epsilon_0} \frac{\frac{1}{2}z\sin\theta}{[d^2 + \frac{1}{4}z^2 - d\cos\theta]^{3/2}}$$

$$M = -F_x \cdot \frac{1}{2}z\sin\theta + F_y \cdot \frac{1}{2}z\cos\theta + F_x \cdot \frac{1}{2}z\sin\theta - F_y \cdot \frac{1}{2}z\cos\theta$$

137 2

$$T\cos\theta - mg = 0$$

$$m\vec{r} = m\vec{r} + T\sin\theta = 0$$



对 $\vec{D}$: qE .

$\vec{D} = -\frac{1}{2} \vec{E}$

01-10

环路定理 $\Leftrightarrow$ 有心力场

证: 有心力 $\vec{F} = f(r) \cdot \vec{r}$

$$\begin{aligned} \text{curl } \vec{F} &= \nabla \times \vec{F} = \nabla \times (f(r) \cdot \vec{r}) \\ &= \nabla f(r) \times \vec{r} + f(r) (\nabla \times \vec{r}) \end{aligned}$$

$$\begin{aligned} \text{curl } \vec{A} &= \left(\frac{1}{r^2 \sin \theta} \left(\frac{\partial(r \sin \theta A_\phi)}{\partial \theta} - \frac{\partial(r A_\theta)}{\partial \phi} \right), \frac{1}{r \sin \theta} \left(\frac{\partial(r \sin \theta A_\phi)}{\partial r} - \frac{\partial A_r}{\partial \phi} \right), \right. \\ &\quad \left. \frac{1}{r} \left(\frac{\partial(r A_\theta)}{\partial r} - \frac{\partial A_r}{\partial \theta} \right) \right) \end{aligned}$$

$$\nabla \times \vec{r} = (0, 0, 0)$$

$$\text{curl } \vec{F} = \nabla f(r) \times \vec{r} = f'(r) \vec{r} \times \vec{r} = 0$$

01-10

例 1. 2. 29

$$G(\theta) = aR \cos \theta = b \cos \theta$$

$$\frac{aR}{d} \cdot \frac{1}{p}$$

先考虑轴线上 $-z$ 点

$$dE_p(\theta) = \frac{1}{4\pi\epsilon_0} \frac{G \cdot 2\lambda R^2 \sin \theta d\theta}{[R^2 \sin^2 \theta + (R \cos \theta - d)^2]^{3/2}} [R \cos \theta - d]$$

$$E_p = \int_0^\pi dE_p = \frac{60R^2}{2\epsilon_0} \int_0^\pi \frac{\sin \theta \cos \theta (d - R \cos \theta) d\theta}{[R^2 + d^2 - 2Rd \cos \theta]^{3/2}}$$

$$\int_0^\pi \frac{\sin \theta \cos \theta d\theta}{[R^2 + d^2 - 2Rd \cos \theta]^{3/2}}$$

$$\int \frac{x dx}{x \sqrt{x}} = -\frac{2(bx + c)}{9 \sqrt{x}}$$

$$Y = \frac{1}{2} \ln \frac{1 + \sqrt{1 - 4x}}{1 - \sqrt{1 - 4x}}$$

$$= \int_{-1}^1 \frac{u \, du}{(R^2 + d^2 - 2Rdu)^{3/2}} = - \frac{2 \left[-2Rdu + 2(R^2 + d^2) \right]}{-(-2Rd)^2 (R^2 + d^2 - 2Rdu)^{1/2}} \Big|_{-1}^1$$

$$= \frac{1}{(-2Rd)^2} \frac{R^2 + d^2 - Rdu}{(R^2 + d^2 - 2Rdu)^{1/2}} \Big|_{u=-1}^1 = \frac{1}{(-2Rd)^2} \left(\frac{R^2 + d^2 - Rd}{R-d} - \frac{R^2 + d^2 + Rd}{R+d} \right)$$

$$= \frac{1}{(Rd)^2} \frac{1}{(R-d)^2} \left[-2R^2d + 2(R^2 + d^2)d \right] = \frac{2d}{R^2(R-d)^2}$$

$$\int_0^{2\pi} \frac{\sin \theta \cos^2 \theta \, d\theta}{(R^2 + d^2 - 2Rd \cos \theta)^{3/2}}$$

$$= \int_{-1}^1 \frac{u^2 \, du}{(R^2 + d^2 - 2Rdu)^{3/2}}$$

$$\int \frac{x \, dx}{x \sqrt{x}} = -\frac{2(bx + 2a)}{9 \sqrt{x}} \quad \int \frac{dx}{x \sqrt{x}} = \frac{2(ax + b)}{9 \sqrt{x}}$$

$$X = a + bx + cx^2 \quad g = 4ac - b^2$$

21-11

电偶极子 $\hookrightarrow$ 求 V 并求 E

$$V = \frac{Q}{4\pi\epsilon_0} \left\{ \left[r^2 + \left(\frac{l}{2}\right)^2 - rl\cos\theta \right]^{1/2} - \left[r^2 + \left(\frac{l}{2}\right)^2 + rl\cos\theta \right]^{1/2} \right\}$$

考虑 $r \gg l$ 的情况

$$\left[r^2 + \left(\frac{l}{2}\right)^2 - rl\cos\theta \right]^{-1/2} = \frac{1}{r} \left[1 - \frac{l}{r}\cos\theta + \frac{l^2}{4r^2} \right]^{-1/2}$$

$$\frac{1}{1+x} \approx 1 - \frac{1}{2}x + \frac{3}{8}x^2$$

$$= \frac{1}{r} \left[1 - \frac{1}{2} \left(-\frac{l}{r}\cos\theta + \frac{l^2}{4r^2} \right) + \frac{3}{8} \left(-\frac{l}{r}\cos\theta + \frac{l^2}{4r^2} \right)^2 \right]$$

舍去 $\frac{l^2}{r^2}$

$$= \frac{1}{r} \left(1 + \frac{1}{2} \frac{l}{r}\cos\theta \right)$$

$$\left[r^2 + \left(\frac{l}{2}\right)^2 + rl\cos\theta \right]^{-1/2} = \frac{1}{r} \left(1 - \frac{1}{2} \frac{l}{r}\cos\theta \right)$$

$$V = \frac{Q}{4\pi\epsilon_0} \frac{l}{r^2} \cos\theta = \frac{Ql\cos\theta}{4\pi\epsilon_0 r^2} = \frac{\vec{p} \cdot \vec{r}}{4\pi\epsilon_0 r^3}$$

$$E = -\nabla V$$

$$= -\frac{\partial}{\partial r} V \vec{r} - \frac{1}{r} \frac{\partial}{\partial \theta} V \vec{\theta}$$

$$= -\frac{1}{4\pi\epsilon_0} \left(\frac{\partial}{\partial r} \frac{p\cos\theta}{r^2} + \frac{1}{r} \frac{\partial}{\partial \theta} \frac{p\cos\theta}{r^2} \vec{\theta} \right)$$

$$= -\frac{1}{4\pi\epsilon_0} \left(-\frac{2p\cos\theta}{r^3} \vec{r} - \frac{p\sin\theta}{r^3} \vec{\theta} \right)$$

$$= \frac{1}{4\pi\epsilon_0 r^3} \left(3\frac{\vec{p} \cdot \vec{r}}{r^2} \vec{r} - \vec{p} \right)$$

電偶極子在外場中的勢能.

$$U = U_+ + U_-$$

$$= q(U_+ - U_-)$$

$$= q \phi \cdot \vec{l}$$

$$= -q \vec{E} \cdot \vec{l} = -\vec{p} \cdot \vec{E}$$

01-11

例 1. 4. 36.

$$\lambda = \frac{q}{4\pi\epsilon_0}$$

$$dZ(r, \theta) = \frac{\lambda dx}{4\pi\epsilon_0} \cdot \frac{1}{r^2 + x^2 - 2rx \cos \theta} \cdot (\cos \theta \vec{r} + \sin \theta \vec{\theta})$$

$$= \frac{\lambda dx}{4\pi\epsilon_0} \cdot \frac{1}{(r^2 + x^2 - 2rx \cos \theta)^{3/2}} [(r - x \cos \theta) \vec{r} + x \sin \theta \cdot \vec{\theta}]$$

$$E(r, \theta) = \int_{-l}^l dZ(r, \theta)$$

$$= \frac{\lambda}{4\pi\epsilon_0} \int_{-l}^l \left[\frac{r dx}{(r^2 + x^2 - 2rx \cos \theta)^{3/2}} \vec{r} + \frac{(\sin \theta \cdot \vec{\theta} - \cos \theta \vec{r}) x dx}{(r^2 + x^2 - 2rx \cos \theta)^{3/2}} \right]$$

$$= \frac{q}{8\pi\epsilon_0 l} \left[r \frac{2(2x - 2r \cos \theta)}{(r^2 + x^2 - 2rx \cos \theta)^{3/2}} \vec{r} \right]_{-l}^l + \frac{(\sin \theta \cdot \vec{\theta} - \cos \theta \vec{r})}{(r^2 + x^2 - 2rx \cos \theta)^{3/2}} \left[\frac{2(-2rx \cos \theta + 2r^2)}{(r^2 + x^2 - 2rx \cos \theta)^{3/2}} \right]_{-l}^l$$

$$= \frac{q}{8\pi\epsilon_0 l} \left[\frac{1}{r \sin^2 \theta} \left(\frac{l - r \cos \theta}{r - l} - \frac{-l - r \cos \theta}{r + l} \right) \vec{r} \right]$$

$$+ (\sin \theta \vec{\theta} - \cos \theta \vec{r}) \frac{1}{r \sin^2 \theta} \left(\frac{r + l \cos \theta}{r + l} - \frac{r - l \cos \theta}{r - l} \right)$$

$$= \frac{q}{8\pi\epsilon_0 l r \sin^2 \theta} (r^2 - l^2) \{ [(r+l)(l - r \cos \theta) + (r-l)(l + r \cos \theta)] \vec{r} \}$$

$$= \frac{q}{8\pi\epsilon_0 l r \sin^2 \theta} (r^2 - l^2) \{ (r+l)(l - r \cos \theta) + (r-l)(l + r \cos \theta) \} \vec{r}$$

$$= \frac{q}{4\pi\epsilon_0 2r\sin^2\theta(r^2-r^2)} [2rL(1-\cos\theta)\hat{r} + (\sin\theta\hat{\theta} - \cos\theta\hat{r}) 2rL(\cos\theta-1)]$$

$$= \frac{q(1-\cos\theta)}{4\pi\epsilon_0 2r\sin^2\theta(r^2-r^2)} [(1-\cos\theta)\hat{r} + \sin\theta\hat{\theta}]$$

$r \gg L$ 时

是点电荷吗?

例 1-11

例 1.4.51



由对称性, 考虑轴线上距中心 d 处一点 P

$$dU_P(\theta) = \frac{q}{4\pi\epsilon_0} \cdot 2\pi R^2 \sin\theta d\theta \cdot \frac{1}{(R^2 + d^2 + 2Rd\cos\theta)^{3/2}}$$

$$U_P = \frac{qR^2}{2\epsilon_0} \int_0^\pi \frac{d(\cos\theta)}{(R^2 + d^2 + 2Rd\cos\theta)^{3/2}} \quad u = \cos\theta$$

$$\int_{-1}^1 \frac{du}{(a+bu)^{3/2}} \quad \text{令 } v = a+bu \quad u = \frac{1}{b}(a-v^2) \quad du = -\frac{2v}{b} dv$$

$$= \int_{a-b}^{a+b} \frac{-\frac{2v}{b} dv}{v^{3/2}} = -\frac{2}{b} \left(\frac{1}{\sqrt{v}} \right) \Big|_{a-b}^{a+b} = -\frac{2}{b} \left(\frac{1}{\sqrt{a+b}} - \frac{1}{\sqrt{a-b}} \right) = -\frac{2}{b} \left(\frac{1}{\sqrt{R+d}} - \frac{1}{\sqrt{R-d}} \right) = -\frac{2}{b} \left(\frac{1}{\sqrt{R+d}} - \frac{1}{\sqrt{R-d}} \right)$$

如果有括号号,

$$U_P = \frac{qR^2}{2\epsilon_0} \cdot \frac{1}{R} = \frac{qR}{2\epsilon_0} = \frac{q}{4\pi\epsilon_0 R}$$

与 d 无关.

例 1-11.

例 1.4.51

$$(1) U_{\text{cm}} = \int_0^\infty \frac{1}{4\pi\epsilon_0} \cdot \frac{\rho_{\text{cm}} \cdot 4\pi r^2 dr}{r}$$

$$= \frac{-q}{4\pi\epsilon_0 a^3} \int_0^\infty e^{-\frac{r}{a}} r^2 dr$$

$$= -\frac{q}{4\pi\epsilon_0 a^3} \left(e^{-\frac{r}{a}} \left(-\frac{r^2}{a} + 2r \right) \Big|_0^\infty - \int_0^\infty e^{-\frac{r}{a}} dr \right)$$

$$= \frac{q}{4\pi\epsilon_0 a^2} \left(-\frac{a^2}{r^3} e^{-\frac{2r}{a}} \right) \Big|_0^\infty = -\frac{q}{4\pi\epsilon_0 a}$$

$$(2) \quad u = u_r + u_\theta = \frac{q}{4\pi\epsilon_0 r} - \frac{q}{4\pi\epsilon_0 a} = 0$$

01-11

例 1-4-60

$$\text{求 } u = \frac{P}{4\pi\epsilon_0 r^2} \cos\theta$$

$$\vec{E} = -\nabla u = \left(-\frac{\partial}{\partial r} u, -\frac{1}{r} \frac{\partial}{\partial \theta} u \right)$$

$$= \left(\frac{2P}{4\pi\epsilon_0 r^3} \cos\theta, \frac{P}{4\pi\epsilon_0 r^2} \frac{\sin\theta}{r} \right)$$

01-13

Gauss 定理 $\rightarrow$ 叠加原理

$$\oint \vec{E} \cdot d\vec{s} = \frac{1}{\epsilon_0} (Q_1 + Q_2)$$

$$= \oint \vec{E}_1 \cdot d\vec{s} + \oint \vec{E}_2 \cdot d\vec{s} = \oint (\vec{E}_1 + \vec{E}_2) \cdot d\vec{s}$$

$$\hookrightarrow \vec{E} = \vec{E}_1 + \vec{E}_2$$

Gauss 定理 + 环路定理 $\rightarrow$ 静电定律



$$\vec{E} = -\nabla u$$

$$\begin{cases} \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \\ \nabla \times \vec{E} = 0 \end{cases} \Rightarrow \nabla^2 u = -\frac{\rho}{\epsilon_0}$$

对于点电荷 Q

$$\rho(r) = Q \delta(r) \quad \nabla^2 u = -\frac{\rho}{\epsilon_0} = -\frac{Q}{\epsilon_0} \delta(r)$$

$$\nabla \cdot \frac{1}{r} = -4\pi \delta(r)$$

$$\nabla^2 u = \frac{Q}{4\pi\epsilon_0 r}$$

$$\rightarrow u = \frac{Q}{4\pi\epsilon_0 r}$$

$$\nabla^2 \frac{1}{r} = -4\pi \delta(r) \text{ for } r \neq 0$$

$$\nabla^2 \frac{1}{r} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + r^2 \sin\theta \frac{\partial}{\partial \theta} \left(\frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \theta} \right) + r^2 \sin^2\theta \frac{\partial^2}{\partial \phi^2}$$

$$\nabla^2 \frac{1}{r} = \begin{cases} 0 & r \neq 0 \\ \infty & r = 0 \end{cases}$$

$$\oint_V f(r) = \lim_{a \rightarrow 0} \iiint_V \nabla^2 \left(\frac{1}{(r^2 + a^2)^{3/2}} \right) dV = \lim_{a \rightarrow 0} f(0)$$

$$f(0) = \iiint_V \frac{1}{r^2} \left(\frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \left(\frac{1}{(r^2 + a^2)^{3/2}} \right) \right) \right) dV$$

$$= \iiint_V \frac{1}{r^2} \frac{\partial}{\partial r} \left(- \frac{r^3}{(r^2 + a^2)^{5/2}} \right) dV$$

$$= \iiint_V \frac{1}{r^2} \left[- \frac{3r^2(r^2 + a^2)^{3/2} - 3r^4(r^2 + a^2)^{1/2}}{(r^2 + a^2)^5} \right] dV$$

$$= \iiint_V \frac{-3a^2}{(r^2 + a^2)^{5/2}} dV$$

$$dV = d\Omega \cdot dr = r^2 dr d\Omega$$

$$\oint(r) = \int_0^{4\pi} d\Omega \int \frac{-3a^2 r^2}{(r^2 + a^2)^{5/2}} dr$$

$$= -12\pi \int \frac{a^2 r^2}{(r^2 + a^2)^{5/2}} dr$$

$$r = a\rho$$

$$= -12\pi \int_0^\infty \frac{a^3 \rho^2}{a^3 (1 + \rho^2)^{5/2}} a d\rho = -12\pi \int_0^\infty \frac{\rho^2}{(1 + \rho^2)^{5/2}} d\rho$$

$$\rho = \tan\theta$$

$$= -12\pi \int_0^{\pi/2} \frac{\tan^2\theta \sec^2\theta}{1 + \tan^2\theta} d\theta = -12\pi \int_0^{\pi/2} \sin^2\theta \cos\theta d\theta = -4\pi$$

0.5

1

01 - 14

21-13

图 1.21

取如图 1.21 所示回路 Γ 。



图 1.22

由

$$W_C = \int_a^b \vec{F}_2 \cdot d\vec{r}$$

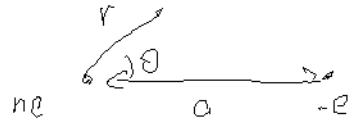
$$\vec{F}_2(\vec{r}) = \frac{q}{4\pi\epsilon_0}$$



Ex 1.24

$$U(r, \theta) = \frac{e}{4\pi\epsilon_0} \left[\frac{n}{r} - \frac{1}{(r^2 + a^2 - 2ar\cos\theta)^{1/2}} \right] \quad n e$$

$$= U_0$$



$$\rightarrow n(r^2 + a^2 - 2ar\cos\theta)^{1/2} - r = \frac{\varphi(r, \theta) U_0}{e} = C$$

$$n^2(r^2 + a^2 - 2ar\cos\theta) = C^2 + r^2 + 2Cr$$

$$(n^2 - 1)r^2 - 2(C + n^2 a \cos\theta)r + n^2 a^2 - C^2 = 0$$

$$r = \frac{1}{2(n^2 - 1)} [4(C + n^2 a \cos\theta)^2 - 4(n^2 - 1)(n^2 a^2 - C^2)]^{1/2}$$

$$= \frac{1}{n^2 - 1} (C^2 + n^4 a^2 \cos^2\theta + 2Cn^2 a \cos\theta - n^4 a^2 - C^2 + n^2 a^2 + n^2 C^2)^{1/2}$$

$$= \frac{1}{n^2 - 1} [n^4 a^2 \sin^2\theta + 2Cn^2 a \cos\theta + n^2(a^2 + C^2)]^{1/2}$$

$$= \frac{n}{n^2 - 1} (-n^2 a^2 \sin^2\theta + 2aC \cos\theta + a^2 + C^2)^{1/2}$$

$U = 0$ ~~at infinity~~: $C = 0$

$$(n^2 - 1)r^2 - 2n^2 a \cos\theta r + n^2 a^2 = 0$$

$$r^2 - \frac{2n^2}{n^2 - 1} a \cos\theta r + \frac{n^4}{(n^2 - 1)^2} a^2 = \frac{n^2}{(n^2 - 1)^2} a^2$$



$$\text{类似 } r^2 + d^2 - 2rd \cos \theta = \text{const.} = r'^2$$

$$d = \frac{n^2}{n^2-1} a \quad r' = \frac{n}{n^2-1} a$$

∴ 为相似曲线。

练习补充

$$\rho \cdot 4\pi r^2 dr = d\phi = d(A \cdot 4\pi r^2)$$

$$\rho = \frac{1}{4\pi r^2} \frac{d\phi}{dr}$$

$$= \frac{1}{4\pi r^2} \frac{d}{dr} (A \cdot 4\pi e^{-br})$$

$$= -A \cdot \frac{b}{r^2} e^{-br}$$